

sol4_fixer_agent_sdk

The working Module 4 loop. A failing branch in. A green one out, or an honest explanation.

`query(), not ClaudeSDKClient.permission_mode: dontAsk.` Merge is never a tool.

Saturday fills two stubs in `labs/lab4_fixer`. Demo `broken-pr` from here.

Read `HOW_TO_RUN.md`, `DESIGN_DOC.md`, and `E2E_PLAN.md`.

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Layout

loop.py	summarize_failure + repair_until_green wrappers
fixer.py	the while loop
doers.py	none / reference / cli / SDK backend
gates.py	pass, retry, escalate
roles.py	dontAsk, one PreToolUse hook, deny tests/**
tests/	hook deny/allow, same-signature escalate

The Deep Agents twin is the graph, not the repair loop.

Setup. Stash first.

```
git -C ../../work/northwind-field-crm stash --include-untracked
cd solutions/sol4_fixer_agent_sdk
echo 'ANTHROPIC_API_KEY=sk-ant-...' >> ../../.env
task setup          # .venv + claude-agent-sdk. PEP 668.
task clone
task reset          # checkout broken-pr. Refuses a dirty clone.
```

Say the stash out loud. That refusal is on brand.

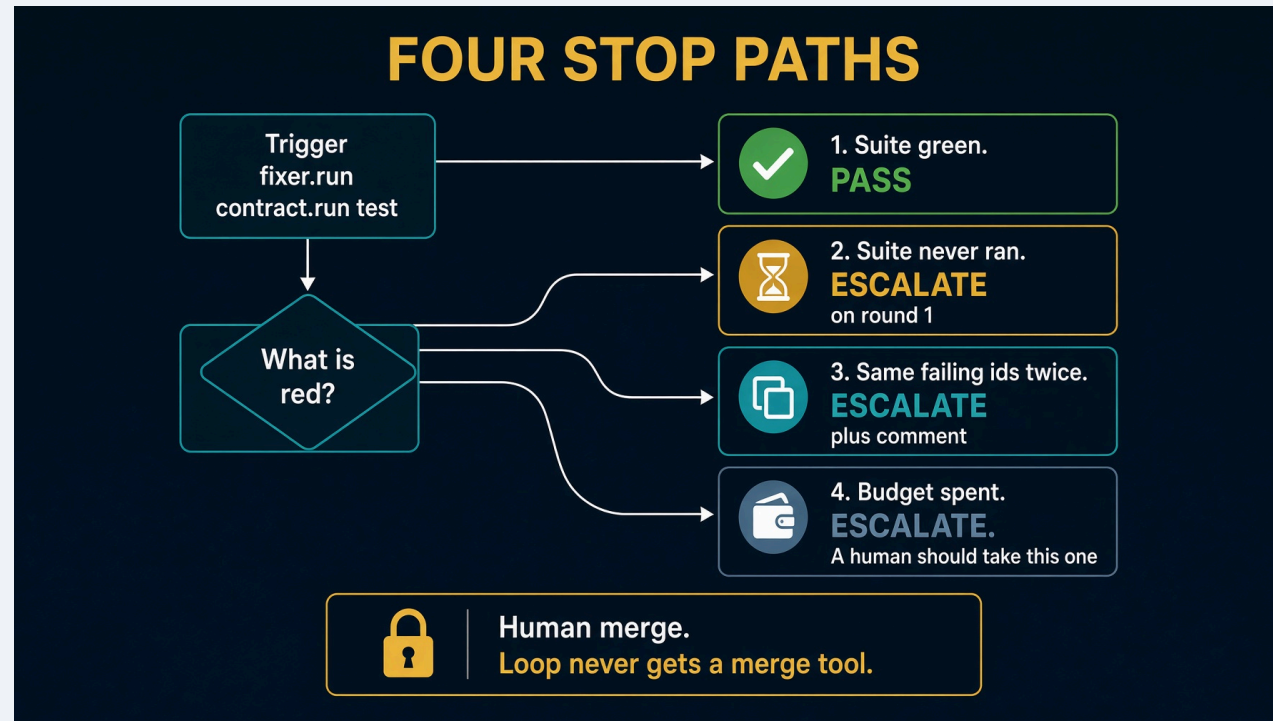
Scripts with no model

```
task table      # judge writes must print no
task test
```

If judge prints `yes`, stop. Neither needs a key, the SDK, or a clone.

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Architecture



Cast is three roles. No planner. See <docs/diagrams/architecture.svg>.

Five unattended lines

DONTASK CONTRACT

-  **1. permission_mode: dontAsk**
never prompts, and fails closed.
-  **2. PreToolUse deny tests/****
cannot weaken a test.
-  **3. maxTurns equals 12**
camelCase. max_turns is a TypeError.
-  **4. Tests after every turn are pytest**
not a claim.
-  **5. Merge is never a tool**
-  **acceptEdits auto-accepts edits before the allow list.**
dontAsk denies anything not pre-approved.

`acceptEdits` auto-accepts edits before the allow list. `dontAsk` denies anything not pre-approved.

Doers

SPEC	NEEDS	BEHAVIOR
none	clone	writes nothing. Loop still reports the truth
reference	clone	copies known-good inside app/**
sdk	task setup + key	AgentSdkBackend wrapping query()

```
task run -- --branch broken-pr --doer none
task run -- --branch broken-pr --doer reference
task run -- --branch broken-pr --doer sdk
```

--research defaults to off. --doer sdk refuses if you skipped task setup.

Expected `--doer none`

```
attempt 1: 1 failing -> retry
attempt 2: 1 failing -> escalate
gate: escalate
The fixer gave up.
A human should take this one.
```

`--doer` reference **against** `broken-pr`: `gate: pass`. Files copied from `known-good` inside `app/**` only.

What Python still owns

`summarize_failure` from `junit`. Four stop paths:

1. Suite green → pass
2. Suite never ran → escalate on round 1
3. Same failing ids twice → escalate, leave a comment
4. Budget spent → escalate, "A human should take this one"

Scope violations escalate even if the suite is green. Spend comes from `total_cost_usd`, so the money gate can fire.

E2E. Disposable clones

Do not use `work/northwind-field-crm` for a live probe. It may hold Lab 2 work.

`E2E_PLAN.md` clones into `/tmp`, prepares the target venv, then runs `--doer none` and `--doer reference` before `--doer sdk`.

`task setup` here installs the SDK. It does not install the CRM's test dependencies.

Troubleshooting

SYMPTOM

Dirty tree

Green on `none`

Edited a test

`acceptEdits` in options

Thought DA repaired it

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FIX

stash, out loud

same ids twice must escalate

full deny envelope on `tests/**`

must be `dontAsk`

that twin is graph only

Recap

Same graph, nobody at the keyboard. Reference doer still bound by WriteScope. Human owns merge.

`dontAsk. Deny tests/**`. Leave a comment when you give up.

The loop is the product. The prompt is not.

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